

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
Operations Research
BCA — Semester 8 — 2024 — Model Answer Sheet
Subject Code: CAOR451

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define Operations Research. Explain the scope and applications of OR in business and industry.

Answer: Operations Research (OR): Scientific approach to decision-making using mathematical models, statistics, and algorithms. Aims to optimize complex systems. **Scope:** Linear programming, network analysis, inventory control, queuing theory, game theory, project management, simulation. **Applications:** **Manufacturing:** Production planning, quality control. **Logistics:** Transportation, supply chain optimization. **Finance:** Portfolio optimization, risk management. **Healthcare:** Scheduling, resource allocation. **Telecommunications:** Network design, traffic routing. **Military:** Strategy optimization. **Process:** Problem formulation → Model construction → Solution → Validation → Implementation.

3. What is linear programming? Formulate a general linear programming problem.

Answer: Linear Programming (LP): Mathematical method for optimizing a linear objective function subject to linear equality and inequality constraints. **General Formulation: Objective:** Maximize or Minimize $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$. **Subject to:** $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$, $a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$, ..., $a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$. **Non-negativity:** $x_1, x_2, \dots, x_n \geq 0$. **Components:** Decision variables, objective function, constraints, non-negativity conditions. **Assumptions:** Linearity, additivity, divisibility, certainty, finite choices.

4. Solve the following LPP graphically: Maximize $Z = 3x + 2y$ subject to $x + y \leq 4$, $x \leq 3$, $y \leq 3$, $x, y \geq 0$.

Answer: Graphical Solution: 1) Plot constraints on x-y plane. $x + y = 4$ passes through (4,0) and (0,4). $x = 3$ is vertical line. $y = 3$ is horizontal line. 2) Feasible region is polygon with vertices: (0,0), (3,0), (3,1), (1,3), (0,3). 3) Evaluate Z at each vertex: $Z(0,0)=0$, $Z(3,0)=9$, $Z(3,1)=11$, $Z(1,3)=9$, $Z(0,3)=6$. 4) Maximum $Z = 11$ at (3,1). **Answer:** $x = 3$, $y = 1$, Maximum $Z = 11$. **Note:** Graphical method works only for 2 variables. For more variables, use Simplex method.

5. Explain the Simplex method for solving linear programming problems.

Answer: Simplex Method: Iterative algorithm for solving LPPs. Developed by George Dantzig. **Steps:** 1) Convert inequalities to equations using slack/surplus/artificial variables. 2) Set up initial simplex tableau with objective function and constraints. 3) Identify pivot column (most negative coefficient in Z-row for

maximization). 4) Identify pivot row using minimum ratio test (RHS/pivot column positive entries). 5) Perform row operations to make pivot element 1 and other elements in column 0. 6) Repeat until all Z-row coefficients are non-negative (optimal for maximization). 7) Read solution from RHS column. **Special Cases:** Degeneracy, alternative optima, unboundedness, infeasibility.

6. What is the transportation problem? Explain the Northwest Corner method for finding initial basic feasible solution.

Answer: Transportation Problem: Special type of LP for minimizing cost of distributing goods from sources to destinations. **Balanced:** Total supply = Total demand. **Northwest Corner Method:** 1) Start at northwest (top-left) cell of transportation table. 2) Allocate maximum possible = $\min(\text{supply}, \text{demand})$. 3) If supply exhausted, move down to next row. If demand satisfied, move right to next column. 4) Repeat until all supplies and demands are satisfied. **Example:** For 3 sources and 4 destinations, requires $3+4-1=6$ basic variables. **Limitation:** Doesn't consider costs; may be far from optimal. **Improvement:** Use Vogel's Approximation or Least Cost method for better initial solutions, then MODI method for optimization.

7. Describe the Critical Path Method (CPM). How is it different from PERT?

Answer: CPM (Critical Path Method): Project management technique for planning and controlling complex projects. **Steps:** 1) List all activities with durations. 2) Determine dependencies. 3) Draw network diagram (AOA or AON). 4) Calculate Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF) for each activity. 5) Calculate slack = $LS - ES$ or $LF - EF$. 6) Critical path is longest path with zero slack activities. **PERT vs CPM:** PERT uses probabilistic time estimates (optimistic, pessimistic, most likely) for research projects. CPM uses deterministic times for repetitive projects. PERT focuses on time; CPM on time-cost tradeoff. Both identify critical path and use network diagrams.

8. What is queuing theory? Explain the basic components of a queuing system.

Answer: Queuing Theory: Mathematical study of waiting lines (queues). Helps optimize service systems by balancing customer wait times against service costs. **Components:** **Arrival Process:** Pattern of customer arrivals (often Poisson distribution with rate λ). **Queue Discipline:** Order of service (FIFO, LIFO, priority). **Service Mechanism:** Number of servers, service rate μ , service time distribution. **System Capacity:** Maximum queue length (finite or infinite). **Population:** Finite or infinite calling population. **Kendall Notation:** A/B/C/D/E where A=arrival, B=service, C=servers, D=capacity, E=population. **Common Model:** M/M/1 (Poisson arrivals, exponential service, 1 server). **Metrics:** L (avg customers in system), W (avg time in system), L_q (avg in queue), W_q (avg wait time).

Group C

Attempt any TWO questions. [$2 \times 10 = 20$]

9. Explain the assignment problem and the Hungarian method for solving it. Solve the following assignment problem to minimize cost:

Person A: [10, 12, 15, 18]

Person B: [14, 11, 13, 16]

Person C: [12, 14, 10, 15]

Person D: [16, 15, 12, 11]

Answer: Assignment Problem: Special LP for assigning n agents to n tasks with minimum cost/time. Each agent does exactly one task. **Hungarian Method:** 1) Row reduction: Subtract minimum of each row from all elements in that row. 2) Column reduction: Subtract minimum of each column. 3) Draw minimum lines to cover all zeros. If lines = n , optimal assignment exists. If not, subtract smallest uncovered element from uncovered elements and add to intersection elements. Repeat. 4) Make assignments at zero positions. **Solution:** After

applying steps: A→Job1(10), B→Job2(11), C→Job3(10), D→Job4(11). Total minimum cost = 10+11+10+11 = 42.

10. What is game theory? Explain two-person zero-sum games. Solve the following game using maximin-minimax principle and check for saddle point:

Player A strategies: [3, -1, 2; -2, 4, 1; 1, 0, 3]

Answer: Game Theory: Study of strategic decision-making among rational players. **Two-Person Zero-Sum Game:** One player's gain equals other's loss. Payoff matrix shows gains for row player (losses for column player). **Maximin-Minimax: Row Player (Maximin):** Find minimum of each row, then maximum of these minima. **Column Player (Minimax):** Find maximum of each column, then minimum of these maxima. **Saddle Point:** If maximin = minimax, game has saddle point (pure strategy). **Solution:** Row minima: [$\min(3, -1, 2) = -1$, $\min(-2, 4, 1) = -2$, $\min(1, 0, 3) = 0$] → Maximin = 0. Column maxima: [$\max(3, -2, 1) = 3$, $\max(-1, 4, 0) = 4$, $\max(2, 1, 3) = 3$] → Minimax = 3. Since $0 \neq 3$, no saddle point. Requires mixed strategy (probability approach).

11. Explain inventory control models. Derive the Economic Order Quantity (EOQ) formula. What are the assumptions of the EOQ model? How does it help in inventory management?

Answer: Inventory Control: Managing stock levels to balance holding costs against ordering costs and stockout risks. **EOQ Model:** Determines optimal order quantity that minimizes total inventory costs. **Total Cost = Ordering Cost + Holding Cost.** $TC = (D/Q)S + (Q/2)H$ where D=annual demand, Q=order quantity, S=ordering cost per order, H=holding cost per unit per year. **Derivation:** Differentiate TC with respect to Q and set to zero: $d(TC)/dQ = -DS/Q^2 + H/2 = 0 \rightarrow Q^2 = 2DS/H \rightarrow EOQ = \sqrt{(2DS/H)}$. **Assumptions:** Constant demand, instant replenishment, no stockouts, constant costs, single item. **Benefits:** Reduces total inventory cost, prevents overstocking/understocking, improves cash flow, aids reorder point decisions.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.